Supplementary Information
Cost–Resilience Thresholds for Battery–Hydrogen Hybrid Microgrids in Islanded Power Systems

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Supplementary file caption

This supplementary file documents the fixed-portfolio definitions, synthetic-load construction, outage-setting rationale, screened parameter ranges, and reproducibility workflow used in the main manuscript.

S1. Fixed portfolio definitions and design intent

Table S1 reports the fixed capacities used in the three candidate portfolios. These values are taken directly from the scenario configuration used to generate the submitted results.

Table S1: Fixed capacities used in the candidate portfolios.

Portfolio	Label	PV (MW)	Battery power (MW)	Battery energy (MWh)	Diesel (MW)	Electrolyzer (MW)	Fuel cell (MW)
battery_only	PV + battery	7.0	2.2	24.0	0.0	0.0	0.0
diesel_battery	PV + diesel + battery	2.8	1.5	8.0	2.4	0.0	0.0
battery_hydrogen	PV + battery + hydrogen	14.0	2.0	12.0	0.0	5.0	2.2

The battery–hydrogen portfolio additionally includes a hydrogen tank with a fixed inventory capacity of 10,000 kg. The fixed-capacity design is intentional: it allows the manuscript to compare operational and resilience behavior across candidate portfolios under common assumptions without claiming endogenous optimal sizing.

S2. Synthetic load archetype and case-study inputs

The case-study inputs were generated from the scenario file and the prepared 8760-hour dataset used by the dispatch model. The synthetic load archetype is calibrated using the following inputs:

- peak demand = 1.65 MW
- load factor = 0.72
- evening peak strength = 0.30
- morning peak strength = 0.10
- commercial load fraction = 0.14
- tourism multiplier = 1.08 in March, April, and December

- seasonal monthly multipliers = [1.02, 1.03, 1.04, 1.05, 1.04, 1.02, 0.99, 0.98, 0.99, 1.00, 1.01, 1.03]
- random noise fraction = 0.025 with seed = 42

The diurnal shape is anchored to local time (UTC+8), so the evening peak falls at local 20:00; the prepared series retains a UTC time index for alignment with the NASA POWER weather data. This construction is not intended to reproduce a licensed feeder's exact dispatch history. Its purpose is to provide a transparent island-like load profile that preserves explicit assumptions about diurnal shape, seasonal variation, commercial demand share, and tourism-sensitive peaks.

S3. Outage-setting rationale and robustness note

The base resilience setting assumes 4 outage events per year, each with 48 h duration, a critical-load fraction of 0.55, and diesel unavailability during the outage windows. This should be interpreted as a severe island contingency stress case that represents fuel-logistics disruption rather than a universal assumption for all outages.

The base resilience results (single representative outage-seed run) are:

- battery_only: CLSR = 1.000, EENS = 0.0 MWh, survivable outage duration = 48 h
- diesel_battery: CLSR = 0.654, EENS = 44.3 MWh, survivable outage duration = 1 h (mean 7.3 h)
- battery_hydrogen: CLSR = 1.000, EENS = 0.0 MWh, survivable outage duration = 48 h

Across ten random outage-start seeds the worst-case diesel–battery outcome is more severe (CLSR $\approx$ 0.58, 0 h survivable); this worst-case value, rather than the single-seed figure above, is the one reported in the main text (Section 4.5, case D0), because a single optimistic seed understates the diesel-unavailable risk.

Reviewer-facing robustness message: the diesel-unavailable setting is used to test whether hydrogen materially changes the resilience envelope under severe local-resource constraints. The paper's claim is therefore conditional and screening-oriented. It does not assert that diesel is unavailable in every real outage, nor that the reported ranking would be identical under all diesel-support assumptions.

S4. Screened parameter ranges and rationale

Table S2 summarizes the exogenous dimensions used for threshold screening.

Table S2: Exogenous dimensions used for threshold screening.

Dimension	Values used	Rationale
Delivered diesel cost	180, 380, 700 USD/MWh	Screens low, base, and high fuel-logistics conditions for remote island supply chains.
Outage duration	12, 48, 96 h	Captures short, severe, and prolonged outage conditions relevant to resilience screening.
Hydrogen CAPEX multiplier	0.45, 0.70, 1.00	Tests how threshold movement responds to lower hydrogen subsystem cost assumptions.
Carbon price	0, 150, 300 USD/tCO_2	Represents a neutral case plus progressively stronger carbon-exposure stress.

At the base outage duration of 48 h and carbon price of 150 USD/tCO_2, the annual cost of battery–hydrogen relative to diesel–battery is positive (more expensive) at low delivered diesel cost and full hydrogen CAPEX, and turns negative (cheaper) as delivered diesel cost rises toward the upper screened value of 700 USD/MWh; the crossover occurs at progressively lower diesel cost as the hydrogen CAPEX multiplier falls to 0.70 or 0.45. This is the threshold logic visualized in the main-text contour (Fig. 2).

S5. Reproducibility workflow and repository-ready contents

The submission package is backed by a repository-ready workflow containing:

- `configs/scenarios.json` for case, cost, resilience, and sensitivity assumptions
- `data/philippines_offgrid_8760.csv` (the 8760-hour model input: real NASA POWER weather and the calibrated UTC+8 load archetype)
- dispatch model modules in `model/`
- reproducible scripts in `scripts/`
- output tables in `results/`
- generated figures in `figures/`

The canonical input dataset is real NASA POWER hourly data (CERES SYN1deg + MERRA-2, year 2025, grid cell 11.0°N 123.0°E, UTC), pre-placed at `data/external/nasa_power_philippines_2025.csv`. The exact Windows commands documented in the repository README are:

Step 1 — data preparation, base scenarios, and base dispatch figure:

```
.\.venv\Scripts\python scripts\prepare_case_study_data.py
.\.venv\Scripts\python scripts\run_scenarios.py
.\.venv\Scripts\python scripts\plot_base_dispatch.py
```

Steps E1–E6 — extended analyses:

```
.\.venv\Scripts\python scripts\run_fine_sweep.py
.\.venv\Scripts\python scripts\plot_fine_sweep.py
.\.venv\Scripts\python scripts\run_cost_driver.py
```

```
.\.venv\Scripts\python scripts\plot_cost_driver.py
.\.venv\Scripts\python scripts\run_matched_backbone.py
.\.venv\Scripts\python scripts\plot_matched_backbone.py
.\.venv\Scripts\python scripts\run_outage_robustness.py
.\.venv\Scripts\python scripts\plot_outage_robustness.py
.\.venv\Scripts\python scripts\run_dynamics.py
.\.venv\Scripts\python scripts\plot_dynamics.py
```

This supplementary file is intended to reduce reviewer ambiguity about portfolio definitions, scenario inputs, and the reproducibility status of the paper. A public repository release can be created upon acceptance without changing the scientific content of the submitted manuscript.

S6. Primary frequency-stability screening: model parameters and regime framing

The reduced-order primary frequency-response model (E6) uses the aggregate swing equation plus first-order droop for each responding source:

$$\frac{d\Delta f}{dt} = \frac{f_0}{2H_{\mathrm{eq}}S_{\mathrm{sys}}}\big[\textstyle\sum_i \delta P_i - \Delta P_{\mathrm{dist}}(t) - D_{\mathrm{eq}}\,\Delta f\big], \qquad \tau_i\,\tfrac{d(\delta P_i)}{dt} = -\delta P_i - K_i\,\Delta f,$$

where $f_0 = 50\,\mathrm{Hz}$, $S_{\mathrm{sys}} = 1.65\,\mathrm{MW}$, and $D_{\mathrm{eq}} = 1.5\,P_{\mathrm{load}}/f_0$. This is a primary-frequency-response screening model only. Sustained power rebalancing after large PV loss or generator trip is provided by secondary control, under-frequency load shedding, and hourly dispatch — none of which are modelled here.

Table S3: Dynamics model parameter values and sources.

Parameter	Value	Unit	Source
H_{bat} (battery VSM virtual inertia)	1.0	s	D'Arco & Suul 2014
H_{dg} (diesel genset real inertia)	2.0	s	Kundur 1994
$\mathrm{Droop}_{\mathrm{bat}}$	5	%	IEEE 1547-2018
$\mathrm{Droop}_{\mathrm{dg}}$	4	%	Kundur 1994
$\mathrm{Droop}_{\mathrm{fc}}$	6	%	Uzunoglu & Alam 2006
τ_{bat} (battery inverter)	0.2	s	Rocabert et al. 2012
τ_{dg} (diesel governor)	2.0	s	CIGRE WG C4.110
τ_{fc} (fuel cell)	5.0	s	Li & Bhatt 2011

The 500 ms windowed RoCoF $= |\Delta f(0.5\,\mathrm{s})|/0.5$ contextualises instantaneous peaks. The diesel–battery diesel-trip case shows 27.5 Hz/s instantaneously but moderates to 5.91 Hz/s when averaged over the first 500 ms, reflecting battery fast-droop recovery ($\tau = 0.2\,\mathrm{s}$); by contrast, evening-peak load-step disturbances stay below 0.5 Hz/s over the same window. The battery–hydrogen portfolio marginally exceeds the 2.0 Hz/s islanded instantaneous-RoCoF threshold at peak load (2.06 Hz/s); tuning the VSM virtual inertia H from 1.0 s to ≥ 1.2 s resolves this as a standard control-parameter choice.

The resilience-adjusted threshold contour (identical layout to the main-text Fig. 2, but using resilience-adjusted annual cost) is provided as Supplementary Fig. S1 (file `figures/fig03b_threshold_contours_resilience_adjusted.pdf`).